Supporting Information

How Far Can Structure-Based Machine Learning Predict Experimental Singlet–Triplet Gaps? An Honest Benchmark for TADF Emitter Triage

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S1. Computational details

S1.1 Semi-empirical calculations

Ground-state geometries were optimised with GFN2-xTB[1] (`xtb` 6.7.1, default convergence: 10^{-6} Ha SCF). Lowest singlet and triplet excitations were obtained with sTDA[2]. These semi-empirical excited states are used only to derive the natural transition orbital (NTO) descriptors of table S1; they are not used as a prediction target (the semi-empirical gap is circular with respect to such features; see main text).

S1.2 DFT reference calculations

Functional-dependence benchmarks (table S10) used B3LYP/6-31G(d) and CAM-B3LYP/def2-TZVP (ORCA 6.1.1[3]), in gas phase and toluene (CPCM).

S2. Natural transition orbital descriptors

NTOs[4] were obtained by singular-value decomposition of the transition density matrix; the dominant hole–electron pair ($\lambda_{\mathrm{NTO}} > 0.9$) defines the spatial descriptors. The 35 NTO descriptors—which require the sTDA excited-state calculation but exclude the S_1/T_1 energy scalars—are enumerated in table S1. For each of the lowest singlet (S_1) and triplet (T_1) states we computed the hole–electron spatial overlap S_{he}, the charge-transfer number CT, the donor/acceptor NTO weights λ_D/λ_A, the hole–electron separation Δr, and the localisation indices (hole-on-acceptor, particle-on-donor). State differences ($\Delta X = X_{S_1} - X_{T_1}$) and their magnitudes $|\Delta X|$ were added because the S_1–T_1 exchange splitting depends on how the spatial distributions *change* between states. The oscillator strength f_{S_1} and the log-transformed absorption proxies are the only intensity-related quantities; no energy scalar enters the feature set.

Table S1 The 35 NTO spatial descriptors used as the structure-derived feature set. X_{S_1} and X_{T_1} denote the quantity evaluated in the singlet and triplet state; $\Delta X = X_{S_1} - X_{T_1}$. Energy scalars ($E_{S_1}$, E_{T_1}, and the gap) are deliberately excluded.

Family	Descriptors
Hole–electron overlap (5)	$S_{\mathrm{he}}(S_1)$, $S_{\mathrm{he}}(T_1)$, ΔS_{he}, $\lvert\Delta S_{\mathrm{he}}\rvert$, $\mathrm{Char}^2_{\mathrm{diff}}$
Charge-transfer number (4)	$\mathrm{CT}(S_1)$, $\mathrm{CT}(T_1)$, $\Delta\mathrm{CT}$, $\lvert\Delta\mathrm{CT}\rvert$
Donor weight (4)	$\lambda_D(S_1)$, $\lambda_D(T_1)$, $\Delta\lambda_D$, $\lvert\Delta\lambda_D\rvert$
Acceptor weight (4)	$\lambda_A(S_1)$, $\lambda_A(T_1)$, $\Delta\lambda_A$, $\lvert\Delta\lambda_A\rvert$
CT separation (4)	$\Delta r(S_1)$, $\Delta r(T_1)$, $\Delta(\Delta r)$, $\lvert\Delta(\Delta r)\rvert$
Localisation (4)	hole-on-A(S_1), particle-on-D(S_1), hole-on-A(T_1), particle-on-D(T_1)
NTO overlap composites (7)	$S^{\mathrm{sum}}_{\mathrm{NTO}}$, $S^{\mathrm{prod}}_{\mathrm{NTO}}$, $S^{\mathrm{ratio}}_{\mathrm{NTO}}$, ΔS_{NTO}, $\lvert\Delta S_{\mathrm{NTO}}\rvert$, NTO overlap($S_1$), NTO overlap($T_1$)
Intensity proxies (3)	f_{S_1}, $\log\lvert A_{S_1}\rvert$, $\log\lvert A_{T_1}\rvert$

S3. Machine learning details

S3.1 Target and dataset

The target is the experimental ΔE_{ST}, extracted from the literature and matched by compound name; multiple reports were reduced to the median. The benchmark contains 231 molecules spanning 212 Bemis–Murcko scaffolds[5]; 92.9 % of scaffolds are singletons, so a scaffold split is numerically close to a random split and the source-paper (DOI-grouped) split is the harder test of transfer.

The corpus was assembled by a name-based join, which is a fragile step; we therefore state the rule and its accounting in full (table S2). Starting from 1 520 extracted literature rows, we kept rows whose property specifier contained “EST” (1 490), parsed the reported value (all 1 490 parsed), and retained values in the physical range $[-0.3, 1.5]$ eV (all 1 490). Each row's compound-name list was exploded and matched by exact string equality to the identifiers of molecules for which we had computed NTO descriptors, giving 363 name–value matches over 231 unique molecules. Of these, 102 molecules carried more than one literature report; we collapsed each to its median. The inter-report spread (standard deviation across reports for a given molecule) had a median of 0 eV but reached 0.564 eV in the worst case, which bounds the achievable accuracy from below. Every matched molecule had an RDKit-parseable SMILES (zero dropped), yielding the final set of 231 molecules over 212 scaffolds. The script `code/si_supplementary_data.py` regenerates this table.

Table S2 Provenance of the experimental ΔE_{ST} corpus: rows retained or dropped at each stage of the name-based join.

Stage	Count
Extracted literature rows	1520
Specifier contains “EST”	1490
Value parsed and finite	1490
Value in physical range $[-0.3, 1.5]$ eV	1490
Name–value matches to descriptor set	363
Unique molecules matched	231
of which with > 1 report	102
Dropped: no RDKit-parseable SMILES	0
Final benchmark molecules	**231**
Final Bemis–Murcko scaffolds	**212**

The corpus records no measurement medium: the solvent and atmosphere fields are empty for all 1 490 EST rows, so the target cannot be stratified by solvent. The label heterogeneity is instead quantified directly from the spread among independent reports of the same molecule—the empirical footprint of the uncontrolled solvent, host and method effects that dominate experimental ΔE_{ST}[6] (table S3). Across the 102 molecules with repeat reports, the inter-report range exceeds the model MAE (0.096 eV) for 19.6 % of molecules, and a single randomly chosen report sits 0.072 eV (RMS) from its molecule's median, while the median itself carries an RMS standard error of 0.049 eV. Label imprecision is thus a material term in the error budget without accounting for all of it; we do not claim it sets a floor the model has reached. Regenerated by `code/solvent_analysis.py`.

Table S3 Label heterogeneity of the experimental target, measured from within-molecule disagreement among independent literature reports (102 molecules with > 1 report). Range is max−min; SD is the standard deviation across a molecule's reports (the latter matches table S2).

Quantity (eV unless noted)	Value
Molecules with > 1 report	102
Inter-report range: mean / median	0.062 / 0.000
Inter-report range: 90th pct. / max	0.17 / 1.09
Fraction with range > model MAE	19.6%
Inter-report SD: mean / max	0.036 / 0.564
Single report vs median (RMS)	0.072
Model MAE (reference)	0.096

S3.1b Feature–target leakage, quantified

Regressing the computed gap $E_{S_1} - E_{T_1}$ on features that contain its two constituents recovers arithmetic rather than structure–property information. We quantified this on the same Bemis–Murcko scaffold `GroupKFold`($k = 5$) used throughout: the target is the sTDA computed gap, and the features are the 35 NTO spatial descriptors with and without the S_1/T_1 energy scalars (table S4). A linear model supplied with the energies recovers the target to floating-point precision. How large the leak appears depends on whether the learner can express a difference of two continuous inputs: a random forest is axis-aligned and cannot, so it reports a far milder R^2 on identical leaked features. Regenerated by `code/a007_leakage_and_nested_cv.py` → `results/a007_leakage_nested_cv.json`.

Table S4 Cross-validated recovery of the computed gap $E_{S_1} - E_{T_1}$, with and without its constituent energies among the features ($n = 231$, scaffold `GroupKFold`, seed 0).

	with S_1/T_1 energies		energies excluded	
Learner	MAE (eV)	R^2	MAE (eV)	R^2
Linear regression	0.000	1.000	0.104	0.36
Ridge	0.018	0.981	0.102	0.38
Random forest	0.081	0.587	0.089	0.52
SVR	0.095	0.411	0.110	0.26

S3.2 Models and evaluation

Model. Random forests[7] (400 trees, scikit-learn[8]) were compared against ridge regression, support-vector regression, gradient boosting, a multilayer perceptron and a mean-value baseline; the random forest was best on every feature set. The forest used 400 estimators with no depth cap, $\sqrt{p}$ features per split, and a fixed seed. Leaving hyperparameters untuned does not by itself remove selection bias, because the *estimator* was chosen using the same folds on which performance is reported. We therefore repeated the comparison under nested cross-validation, selecting the learner inside each training fold and scoring it on the untouched outer fold: the random forest is selected in 5/5 outer folds and the nested MAE equals the reported value to five decimal places (0.095 63 eV), so the selection optimism is 0 eV (`code/a007_leakage_and_nested_cv.py`).

Cross-validation. Performance is reported under Bemis–Murcko scaffold `GroupKFold` ($k = 5$): each fold holds out whole scaffolds, so no scaffold appears in both training and test, which removes the optimistic leakage of a random split on a set where 212 scaffolds cover 231 molecules.

Intervals and importance. 95 % confidence intervals come from 2 000 bootstrap resamples of the pooled out-of-fold predictions (table S5). Feature importance used exact TreeSHAP[9].

Table S5 Prediction of experimental ΔE_{ST} (random forest, scaffold cross-validation, 231 molecules); R^2 with 95 % bootstrap CI. The 2D-structure-only sets (no quantum chemistry) match the NTO set (which needs an sTDA run); structure-only R^2 intervals include zero.

Feature set	MAE (eV)	R^2 (95% CI)	ρ
Morgan FP (2D structure only)	0.091	0.27 (−0.19, 0.53)	0.31
RDKit desc. (2D structure only)	0.100	0.31 (−0.11, 0.51)	0.26
NTO spatial (needs sTDA)	0.096	0.25 (0.03, 0.45)	0.36
Energies + NTO (reference)	0.097	0.24	0.34
Mean-value baseline	0.107	0.00	—
Raw sTDA gap	0.251	−2.22	0.24

Table S6 collects all accuracy and association metrics for the headline NTO model with their 95 % bootstrap confidence intervals, together with the power analysis for the rank-correlation test. Both Spearman ρ and Pearson r exclude zero, while the R^2 interval does not; the test for $\rho = 0.36$ is fully powered at $n = 231$ (only $n \approx 60$ is required for 80 % power), so the wide R^2 interval is a property of the variance-explained statistic under outlier influence rather than a sample-size limitation. Values regenerated by `code/r2_ci_power_analysis.py`.

Table S6 Headline NTO random-forest model: accuracy and association metrics with 95 % percentile-bootstrap CIs (5 000 resamples of the out-of-fold predictions, 231 molecules), and a Fisher-z power analysis for the correlation test (two-sided, $\alpha = 0.05$).

Metric	Point estimate	95% CI
MAE (eV)	0.096	(0.082, 0.109)
R^2	0.25	(0.03, 0.45)
Spearman ρ	0.36	(0.24, 0.47)
Pearson r	0.50	(0.26, 0.68)
Power analysis (correlation test, $\rho = 0.36$)		
Spearman ρ p-value	2.6×10^{-8}	
Achieved power at $n = 231$	≈ 1.0	
n for 80 % power	60	
n for 90 % power	79	

A scaffold-CV learning curve (fig. S1) tests whether the 231-molecule set is simply too small. The cross-validated MAE shows no systematic decrease beyond $n \approx 90$: it varies by 5 % across $n = 46$–231, within the fold-to-fold standard deviation (≈ 0.013 eV), while the training MAE stays near 0.035 eV. The persistent train–CV gap that does *not* close with more data shows the model is not merely undersampled. The curve is not formally a plateau (`results/learning_curve.json` records `"plateau": false`) and the training sizes run only to $n = 184$; within this data regime, additional molecules of the same provenance did not lower the error. We do not read the gap as identifying an irreducible label-noise floor, since a train–CV gap is equally consistent with misspecification. Regenerated by `code/learning_curve_analysis.py`.

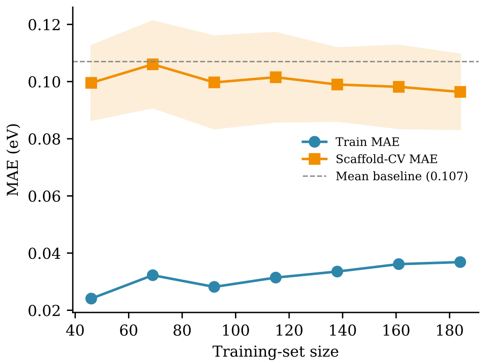

Figure S1 Scaffold cross-validation learning curve for the headline NTO random forest. Shaded band: ± 1 fold standard deviation of the CV-MAE. The CV-MAE is flat beyond $n \approx 90$ and remains above the mean-value baseline gap, showing the error does not fall with more data of the same provenance.

S3.3 Per-feature SHAP attributions

Table S7 lists the mean absolute TreeSHAP value of every NTO descriptor for the random forest trained on the 231-molecule set. Attribution concentrates on a handful of descriptors: the singlet hole–electron overlap $S_{\mathrm{he}}(S_1)$ alone accounts for 29 % of the total attribution, and the top five descriptors ($S_{\mathrm{he}}(S_1)$, $S_{\mathrm{NTO}}^{\mathrm{ratio}}$, f_{S_1}, $\Delta r(S_1)$, $S_{\mathrm{he}}(T_1)$) for 60 %. The 12 lowest-ranked descriptors together contribute under 3 %; the two raw NTO-overlap and the squared-character features are effectively unused. This is consistent with the main-text finding that the gap is governed by a small number of donor–acceptor overlap quantities rather than by the full 35-dimensional descriptor vector.

The interpretive value of the physics-informed descriptors is visible when their attribution is placed beside that of the equally accurate Morgan-fingerprint model (fig. S2). The ten highest-attribution NTO descriptors carry 79 % of the total and are all named physical quantities; the ten highest-attribution Morgan bits carry only 30 %, spread over 2 048 anonymous hashed substructure bits that resist mechanistic reading. Equal predictive accuracy thus buys interpretability only in the NTO representation. Regenerated by `code/shap_morgan_vs_nto.py`.

To test whether this attribution pattern is a stable signal rather than an artefact of one particular fit, we refit the random forest on 200 bootstrap resamples of the 231-molecule set and recorded the top-five descriptors of each (table S8). The leading descriptor $S_{\mathrm{he}}(S_1)$ appears in the top five of 99.5 % of resamples; on average 3.7 of the five canonical descriptors recur per resample, and 64.5 % of resamples recover at least four of the five. The attribution is therefore reproducible at the top of the ranking, even though the model's ranking power on the gap itself is modest; we accordingly read these attributions as mechanistic hypotheses for experimental test, not validated causal claims. Regenerated by

Table S7 Per-feature importance (mean |SHAP|, in meV) of the 35 NTO descriptors for the random forest on the 231-molecule experimental set, ranked high to low. Regenerated by `code/si_supplementary_data.py`.

Descriptor	meV	Descriptor	meV
$S_{\mathrm{he}}(S_1)$	36.9	$\|\Delta\mathrm{CT}\|$	1.6
$S_{\mathrm{NTO}}^{\mathrm{ratio}}$	14.5	hole-on-A(T_1)	1.5
f_{S_1}	11.5	$\Delta r(T_1)$	1.5
$\Delta r(S_1)$	7.9	$\mathrm{CT}(S_1)$	1.4
$S_{\mathrm{he}}(T_1)$	6.9	$\|\Delta\lambda_A\|$	1.3
$\log\|A_{T_1}\|$	6.9	$\Delta\mathrm{CT}$	1.2
$\Delta(\Delta r)$	4.8	particle-on-D(S_1)	0.7
ΔS_{he}	4.3	$\lambda_D(S_1)$	0.7
$\|\Delta(\Delta r)\|$	4.1	particle-on-D(T_1)	0.6
$\|\Delta S_{\mathrm{he}}\|$	4.0	$\lambda_A(T_1)$	0.6
$\|\Delta\lambda_D\|$	3.1	$\lambda_A(S_1)$	0.5
$\log\|A_{S_1}\|$	3.0	hole-on-A(S_1)	0.5
$\Delta\lambda_D$	2.6	ΔS_{NTO}	0.1
$\Delta\lambda_A$	2.0	$S_{\mathrm{NTO}}^{\mathrm{sum}}$	0.1
$\lambda_D(T_1)$	2.0	NTO overlap(T_1)	0.1
$\mathrm{CT}(T_1)$	1.7	NTO overlap(S_1)	0.0
		$\|\Delta S_{\mathrm{NTO}}\|$	0.0
		$S_{\mathrm{NTO}}^{\mathrm{prod}}$	0.0
		$\mathrm{Char}_{\mathrm{diff}}^2$	0.0

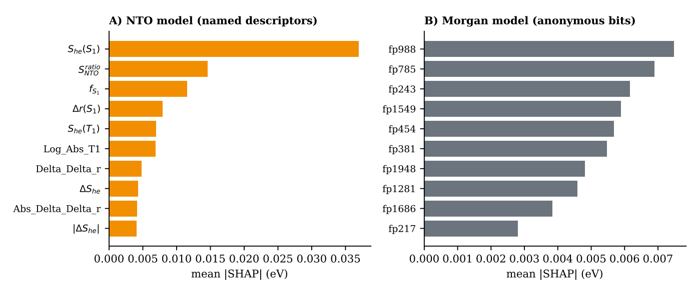

Figure S2 Top-ten SHAP attributions for the two equally accurate models. (A) NTO random forest: attribution concentrates on named donor–acceptor overlap and charge-transfer descriptors. (B) Morgan random forest: attribution is diffuse across anonymous fingerprint bits (fp$n$), which carry no direct chemical meaning.

`code/shap_bootstrap_stability.py`.

Table S8 Bootstrap stability of the top-five SHAP descriptors: fraction of 200 resampled-and-refit models in which each canonical descriptor remains in the top five.

Descriptor	Top-5 frequency
$S_{\mathrm{he}}(S_1)$	99.5%
$S_{\mathrm{NTO}}^{\mathrm{ratio}}$	74.0%
f_{S_1}	72.0%
$\Delta r(S_1)$	68.0%
$S_{\mathrm{he}}(T_1)$	59.5%
Mean top-5 overlap	3.7 / 5
Resamples recovering ≥ 4	64.5%

S3.4 Scaffold-split characterisation

Because 212 Bemis–Murcko scaffolds cover 231 molecules, 92.9 % of scaffolds are singletons, and a reviewer may reasonably ask whether scaffold `GroupKFold` differs at all from a random split. It does, in the way that matters for leakage: by construction no scaffold appears in both training and test (zero shared groups in every fold), whereas a random split of the same data leaks 3 to 8 scaffolds per fold across the train/test boundary (table S9). The numerical effect on difficulty is nonetheless small—the mean nearest-neighbour Tanimoto similarity between a test molecule and its closest training molecule falls only from 0.615 (random) to 0.592 (scaffold)—so we do not claim the split makes the task dramatically harder; we claim it removes a specific leakage channel.

That the model learns chemical signal rather than scaffold-group artefacts is established by a label-permutation test: the target was shuffled 1 000 times and the scaffold-CV MAE recomputed under each permutation. The true MAE (0.095 6 eV) lies far in the left tail of the null distribution (mean 0.121 eV, SD 0.004 3 eV, 95 % interval 0.112 to 0.129), giving $p = 0.001$. Regenerated by `code/scaffold_split_analysis.py`.

Table S9 Scaffold `GroupKFold` versus a random 5-fold split on the same 231 molecules. Scaffold splitting removes train/test scaffold overlap; the label-permutation test confirms the fitted model is far from the no-signal null.

Quantity	Scaffold split	Random split
Shared scaffolds per fold	0 (all folds)	3–8
Test–train NN Tanimoto (mean)	0.592	0.615
Label-permutation test (1000 shuffles, scaffold CV)		
True CV-MAE (eV)	0.0956	
Null CV-MAE: mean (95% CI) (eV)	0.121 (0.112, 0.129)	
p-value (left tail)	0.001	

S3.5 Reproducibility

The scripts `code/finalize_model.py` and `code/rebuild_structural_model.py` regenerate table S5 and Fig. 1 of the main text from the deposited data.

S4. DFT level benchmark

Across 7 representative emitters in gas and toluene, B3LYP/6-31G(d) and CAM-B3LYP/def2-TZVP differ in ΔE_{ST} by a mean absolute 0.30 eV (up to 0.58 eV), with sign reversals of the relative ordering (e.g. DMAC-DPS). This functional dependence sets a physical ceiling on the meaning of any sub-0.1 eV accuracy claim (table S10).

Table S10 Functional dependence of ΔE_{ST} (eV): B3LYP/6-31G(d) vs CAM-B3LYP/def2-TZVP.

Molecule	Phase	B3LYP	CAM-B3LYP	Δ
4CzIPN	gas	0.192	0.361	+0.169
4CzIPN	toluene	0.206	0.335	+0.129
ACRFLCN	gas	0.007	0.590	+0.583
ACRFLCN	toluene	0.000	0.548	+0.548
ACRSA	gas	0.034	0.447	+0.413
ACRSA	toluene	0.000	0.372	+0.372
BACN	gas	0.524	0.991	+0.467
BACN	toluene	0.521	0.921	+0.400
DMAC-DPS	gas	0.489	0.188	−0.301
DMAC-DPS	toluene	0.444	0.207	−0.237
DMAC-TRZ	gas	0.098	0.261	+0.163
DMAC-TRZ	toluene	0.085	0.271	+0.186
PXZ-NAI	gas	0.245	0.355	+0.110
PXZ-NAI	toluene	0.289	0.360	+0.071

A higher-tier check confirms this functional dependence is not an artefact of the disparate B3LYP/CAM-B3LYP pair. Across six range-separated and hybrid functionals applied to 27 charge-transfer emitters[10], the singlet–triplet gap *estimate* ($E(\mathrm{S}_1^{\mathrm{ROKS}}) - E(\mathrm{S}_1^{\mathrm{UKS}}) \approx \Delta E_{\mathrm{ST}}/2$) disagrees per molecule by a median of 0.049 eV (75th percentile 0.10 eV, one outlier excluded). Even among modern functionals the estimate varies at the 0.05 eV to 0.10 eV level; no absolute gap or experimental agreement is claimed from these data. Extracted by `code/parse_roks_benchmark.py` from `results/theoexp_roks_energies.csv`.

Three independently measured noise sources are relevant to the accuracy ceiling, but they are of different kinds and bound different quantities (table S11); we do *not* combine them into a single floor, because only one of them applies to a model trained on the measured gap. The experimental label noise (A, 0.072 eV RMS) is the source that applies to regression on the *measured* target. The relevant figure is not that single-report scatter, however, but the precision of the quantity actually regressed on: the per-molecule median carries an RMS standard error of 0.049 eV, about half the observed random-forest MAE of 0.096 eV (95 % CI 0.082 to 0.109). Label imprecision is therefore a substantial term in the error budget without accounting for all of it, and we do not claim the learner sits at a noise floor. The vertical→adiabatic descriptor offset (B, 0.195 eV MAE) is dominated by an additive, largely rank-preserving shift (S4.1; residual RMS 0.245 eV, $n = 14$) and is absent from the 2D-structure-only models that reach the same accuracy, so it does not floor the experimental target. The functional dependence of the computed reference (C, mean 0.296 eV) bounds how closely any *computed* gap can match experiment—the reason sub-0.05 eV agreement against a computed reference is not physically meaningful—but it likewise does not floor a model regressed on experimental labels. Only the label term applies to the present regression, and it is about half the model error, so we describe the model as label-limited in part rather than as sitting at a ceiling. What the capacity test does establish is the negative half of that statement: no higher-capacity learner improves on it (table S18). Component magnitudes computed by `code/ceiling_decomposition.py`.

Table S11 Three independently measured noise sources relevant to the accuracy ceiling and the quantity each bounds. Only the experimental label term (A) applies to a model trained on the measured gap, and the precision of the median actually regressed on (0.049 eV) is about half the random-forest MAE; B and C bound the descriptor approximation and the computed-reference agreement respectively, and are not combined into a single floor.

	Source (what it bounds)	Magnitude (eV)
A	Single-report scatter about the median — applies to the experimental target (RMS)	0.072
B	Vertical→adiabatic descriptor offset — largely additive, ranking mostly preserved ($n = 14$); not a target floor (MAE)	0.195
C	Functional dependence of the computed reference — bounds computed-gap agreement only (mean)	0.296
	Observed random-forest MAE	0.096
	Mean-value baseline	0.107

S4.1 Vertical versus adiabatic gap

The NTO descriptors are evaluated at the ground-state (vertical) geometry. To bound the error this introduces relative to a relaxed (adiabatic) treatment, we recomputed ΔE_{ST} for 14 representative emitters in gas and toluene at CAM-B3LYP/def2-TZVP (D3BJ, RIJCOSX) with excited-state geometry optimisation in ORCA, and compared with the vertical sTDA-xTB values (table S12). Over the 27 converged points the vertical gap deviates from the adiabatic reference by a mean absolute 0.195 eV (RMSE 0.245 eV, worst case 0.548 eV for 2CzTPE). The deviation is dominated by an additive offset rather than by reordering: a linear fit of adiabatic on vertical gives slope 1.00 and intercept 0.07 eV ($R^2 = 0.70$), with rank correlation Spearman $\rho = 0.71$. The residual is not negligible (RMS 0.245 eV, signed deviations of both signs) and $n = 14$ cannot settle how much of it is systematic. Geometry relaxation therefore shifts the absolute scale while leaving the ordering—on which the triage use depends—largely intact. Regenerated by `code/adiabatic_validation_analysis.py` from `results/vertical_vs_adiabatic_comparison.csv`.

Table S12 Vertical (sTDA-xTB, ground-state geometry) versus adiabatic (CAM-B3LYP/def2-TZVP, excited-state optimisation) ΔE_{ST} (eV) for 14 emitters. Δ = adiabatic − vertical. Summary: MAE 0.195 eV, slope 1.00, intercept 0.07 eV, $R^2 = 0.70$, $\rho = 0.71$.

Molecule	Phase	Vertical	Adiabatic	Δ
4CzIPN	gas	0.361	0.402	+0.041
4CzIPN	toluene	0.335	0.436	+0.101
DMAC-TRZ	gas	0.261	0.442	+0.181
DMAC-TRZ	toluene	0.271	0.638	+0.367
DMAC-DPS	gas	0.188	0.385	+0.197
DMAC-DPS	toluene	0.207	0.558	+0.351
PXZ-NAI	gas	0.355	0.243	−0.112
PXZ-NAI	toluene	0.360	0.491	+0.131
TPA-APy	gas	0.876	0.753	−0.123
TPA-APy	toluene	0.765	0.895	+0.130
BACN	gas	0.991	1.129	+0.138
BACN	toluene	0.921	1.063	+0.142
BMZ-TZ	gas	0.938	0.410	−0.528
BMZ-TZ	toluene	0.933	0.700	−0.233
2CzPN	gas	0.613	0.560	−0.053
2CzPN	toluene	0.621	0.652	+0.031
APPT-PXZ	gas	0.350	0.331	−0.019
2PXZP	gas	0.467	0.384	−0.083
2PXZP	toluene	0.485	0.639	+0.154
ACRSA	gas	0.447	0.204	−0.243
ACRSA	toluene	0.372	0.677	+0.305
ACRFLCN	gas	0.590	0.302	−0.288
ACRFLCN	toluene	0.548	0.576	+0.028
2CzTPE	gas	1.143	1.691	+0.548
2CzTPE	toluene	1.042	1.535	+0.493
BBPA	gas	1.521	1.641	+0.120
BBPA	toluene	1.433	1.562	+0.129

S4.2 Triplet manifold (T1–T2)

A small S_1–T_1 gap does not guarantee fast reverse intersystem crossing; the T_1–T_2 splitting and the wider triplet manifold matter too[11]. The same vertical CAM-B3LYP/def2-TZVP TD-DFT calculations that supplied the adiabatic benchmark also resolve ten triplet roots, so we report T_1 and T_2 directly (table S13). Across the 27 converged emitter–phase points the T_1–T_2 gap ranges from 0.018 eV (4CzIPN) to 1.50 eV (BBPA), mean 0.41 eV; 13 of 27 fall below 0.3 eV. The manifold therefore varies widely *among molecules with similar* ΔE_{ST} (e.g. DMAC-DPS and DMAC-TRZ have near-equal S_1–T_1 gaps but T_1–T_2 of 0.06 eV to 0.26 eV), which is exactly the information a scalar ΔE_{ST} triage cannot encode and a direction where the excited-state pipeline adds genuine value. Regenerated by `code/extract_triplet_manifold.py` from the `orca_package` outputs.

To read the manifold across the whole corpus rather than the 14-molecule ORCA subset, we extracted T_1 and T_2 for all 231 emitters from the cheaper sTDA-xTB pipeline (`code/extract_t1_t2_rerun.py`; 231/231 converged, `results/t1_t2_full.csv`). The full-corpus T_1–T_2 gap has me-

Table S13 Vertical CAM-B3LYP/def2-TZVP TDA energies (eV): S_1, T_1, T_2, the singlet–triplet gap $\Delta E_{\mathrm{ST}} = S_1 - T_1$, and the triplet-manifold gap T_1–T_2, for 14 emitters in gas and toluene.

Molecule	Phase	S_1	T_1	T_2	ΔE_{ST}	T_1–T_2
2CzPN	gas	3.652	3.039	3.464	0.613	0.425
2CzPN	toluene	3.632	3.011	3.459	0.621	0.448
2CzTPE	gas	3.496	2.353	2.767	1.143	0.414
2CzTPE	toluene	3.411	2.369	2.788	1.042	0.419
2PXZP	gas	3.292	2.825	3.080	0.467	0.255
2PXZP	toluene	3.343	2.858	3.139	0.485	0.281
4CzIPN	gas	3.377	3.016	3.050	0.361	0.034
4CzIPN	toluene	3.343	3.008	3.026	0.335	0.018
ACRFLCN	gas	3.597	3.007	3.548	0.590	0.541
ACRFLCN	toluene	3.546	2.998	3.536	0.548	0.538
ACRSA	gas	3.769	3.322	3.513	0.447	0.191
ACRSA	toluene	3.746	3.374	3.500	0.372	0.126
APPT-PXZ	gas	2.701	2.351	2.734	0.350	0.383
BACN	gas	3.713	2.722	3.533	0.991	0.811
BACN	toluene	3.629	2.708	3.503	0.921	0.795
BBPA	gas	3.553	2.032	3.525	1.521	1.493
BBPA	toluene	3.458	2.025	3.526	1.433	1.501
BMZ-TZ	gas	3.730	2.792	3.180	0.938	0.388
BMZ-TZ	toluene	3.896	2.963	3.142	0.933	0.179
DMAC-DPS	gas	3.637	3.449	3.508	0.188	0.059
DMAC-DPS	toluene	3.678	3.471	3.538	0.207	0.067
DMAC-TRZ	gas	3.509	3.248	3.504	0.261	0.256
DMAC-TRZ	toluene	3.533	3.262	3.509	0.271	0.247
PXZ-NAI	gas	2.688	2.333	2.836	0.355	0.503
PXZ-NAI	toluene	2.650	2.290	2.829	0.360	0.539
TPA-APy	gas	3.555	2.679	2.823	0.876	0.144
TPA-APy	toluene	3.425	2.660	2.795	0.765	0.135

dian 0.14 eV (mean 0.24 eV), and 74 % of emitters fall below the 0.3 eV region where a near-degenerate second triplet can open competing non-radiative channels (fig. S3). Most low-gap candidates therefore remain exposed to inefficient reverse intersystem crossing—a hazard a scalar S_1–T_1 triage cannot see.

This full-corpus manifold must be read *qualitatively only*. Benchmarking the sTDA-xTB T_1–T_2 gap against the CAM-B3LYP reference on the six overlapping molecules gives a mean absolute deviation of 0.237 eV (Spearman $\rho = 0.43$, $p = 0.40$; table S14), so the two methods agree on neither magnitude nor rank at a useful level. Quantitative ranking on RISC propensity still requires the higher-level calculation; the sTDA manifold serves only to flag that the triplet-manifold hazard is widespread, not to order candidates by it.

Table S14 sTDA-xTB versus CAM-B3LYP T_1–T_2 gap (eV) for the six molecules present in both sets. Independent methods; the comparison is genuine (fresh GFN2 geometries). Summary: MAE 0.237 eV, Spearman $\rho = 0.43$ ($p = 0.40$, $n = 6$).

Molecule	sTDA-xTB	CAM-B3LYP
2CzPN	0.169	0.425
2PXZP	0.004	0.255
4CzIPN	0.073	0.034
ACRSA	0.006	0.191
DMAC-DPS	0.011	0.059
PXZ-NAI	1.146	0.503

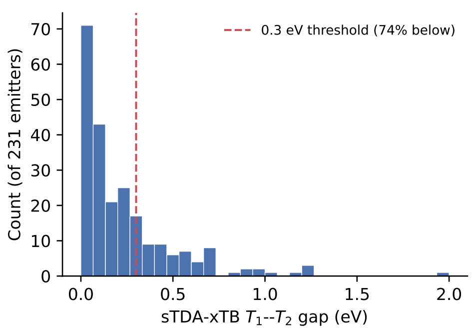

Figure S3 Distribution of the sTDA-xTB T_1–T_2 gap across all 231 emitters. 74 % fall below 0.3 eV (dashed line). Read qualitatively only (table S14).

S4.3 Range-separated TD-DFT reference gap

Referee comment on the original submission held that the semi-empirical sTDA gap understates what a proper range-separated TD-DFT calculation would achieve. We tested this directly. Vertical ΔE_{ST} was computed for all 231 benchmark molecules at wB97X-D4/def2-SVP with RIJCOSX and the Tamm–Dancoff approximation (ORCA 6[3]; `nroots` 8, triplets included), at the GFN2-xTB ground-state geometry, with $\Delta E_{\mathrm{ST}} = E(S_1) - E(T_1)$. All 231/231 jobs terminated normally with all states parsed and no inverted gaps; the run cost 609 core-hours (measured: 76.2 h elapsed on 8 processes; median 16.1 min per molecule, slowest 2.1 h; wall $\propto$ atoms$^{1.89}$). Per-molecule gaps and timings are in `results/tddft_reference_231.csv`; the head-to-head analysis is `results/ref1_tddft_gap.json`, produced by `code/ref1_tddft_gap.py` on the same experimental labels, SMILES, NTO features and scaffold `GroupKFold` folds as table S5.

The raw TD-DFT gap overestimates the measured gap by a systematic 0.654 eV (TD-DFT mean 0.803 eV, SD 0.248; experiment mean 0.149 eV, SD 0.167), more than triple the 0.221 eV sTDA offset. sTDA and TD-DFT gaps agree only moderately with each other ($r = 0.461$, $\rho = 0.402$, mean shift 0.433 eV). An in-fold degree-1 calibration ($a\,g + b$, fitted per training fold on $\approx$185 molecules, no test leak) removes the offset. table S15 lists every predictor on the common folds; table S16 the paired MAE differences (2000-resample bootstrap on identical folds plus Wilcoxon signed-rank on paired absolute errors).

Table S15 TD-DFT reference gap versus semi-empirical gap and structure-only random forests, all on the same 231-molecule scaffold `GroupKFold`($k = 5$) folds. B rows are single-quantity predictors; M rows are random forests (400 trees). "Calibrated" = in-fold linear fit requiring experimental training labels.

	Predictor	MAE /eV	RMSE /eV	R^2	r
B0	Mean baseline	0.107	0.00.167	−0.01	−0.11
B1	Raw sTDA gap	0.251	0.299	−2.22	0.32
B3	Raw TD-DFT gap	0.654	0.694	−16.3	0.43
B2	Calibrated sTDA gap	0.104	0.160	0.08	0.29
B4	Calibrated TD-DFT gap	0.099	0.152	0.17	0.41
M1	RF Morgan (structure only)	0.091	0.142	0.27	0.55
M2	RF NTO (energy-free)	0.096	0.144	0.25	0.50
M5	RF NTO + TD-DFT gap	0.092	0.140	0.29	0.54
M6	RF Morgan + TD-DFT gap	0.087	0.138	0.31	0.58
M7	RF NTO + sTDA + TD-DFT	0.093	0.141	0.29	0.54

Table S16 Paired MAE difference on identical folds (positive = the TD-DFT gap helps). CI from 2 000 paired bootstrap resamples; p from the Wilcoxon signed-rank test on paired absolute errors.

Comparison	ΔMAE /eV	95% CI /eV	Wilc
M6 vs M1 (Morgan + gap)	+0.0039	[+0.0004, +0.0071]	0.
M5 vs M2 (NTO + gap)	+0.0032	[+0.0008, +0.0058]	0.
B4 vs B0 (calib. gap vs mean)	+0.0085	[+0.0002, +0.0164]	0.
B4 vs M1 (calib. gap vs RF)	−0.0075	[−0.0205, +0.0061]	0
B3 vs B1 (raw TD-DFT vs sTDA)	−0.403	[−0.431, −0.375]	1 ×

Read raw, a higher level of theory is worse, not better: the range-separated TDA gap is the least accurate predictor in the study, driven by a large fixed exchange-splitting overestimate. It does track experiment better in rank (r 0.32 to 0.43 from sTDA to TD-DFT) and, once calibrated on experimental labels, in MAE (0.099 eV, below the mean baseline but not the structure-only forest). Added as one random-forest feature it gives the only physics contribution in this work that clears a paired test (M6 vs M1, M5 vs M2), a 4 % MAE reduction for 609 core-hours. Caveats: gas-phase vertical values only (no solvent, no excited-state relaxation); the 0.654 eV bias is specific to wB97X-D4/def2-SVP/TDA; def2-SVP is a small basis, though the systematic offset is expected to persist at a larger basis for a range-separated hybrid; TDA raises triplet energies relative to full TD-DFT, widening the gap, and was the deliberate stable choice; the M5/M6 gains ($\approx$0.003–0.004 eV) are real by the paired test but small against the 0.072 eV single-report label scatter.

S5. Substructure-filtered extended library

An extended set of 22 194 structures was obtained by screening the PubChem CID–SMILES collection for compounds carrying both a donor motif (carbazole, phenothiazine, phenoxazine, acridine, diphenylamine, triphenylamine or dimethylaniline) and an acceptor motif (triazine, benzonitrile, pyridine, pyrimidine, quinoxaline, benzothiadiazole, naphthoquinone or a cyano group), under the constraints MW $< 900\,\mathrm{Da}$, ≥ 2 aromatic rings and ≥ 1 N/O/S heteroatom. It is a substructure filter over an existing database, not a combinatorial enumeration of fragment pairs; see the Methods of the main text. Candidates were ranked by the rule-based TADF score defined in the main text. These are synthesis-planning leads only: they are not quantum-validated, and the heuristic score is not a calibrated ΔE_{ST} predictor. Property distributions of the filtered set versus the top-scoring subset are shown in fig. S4.

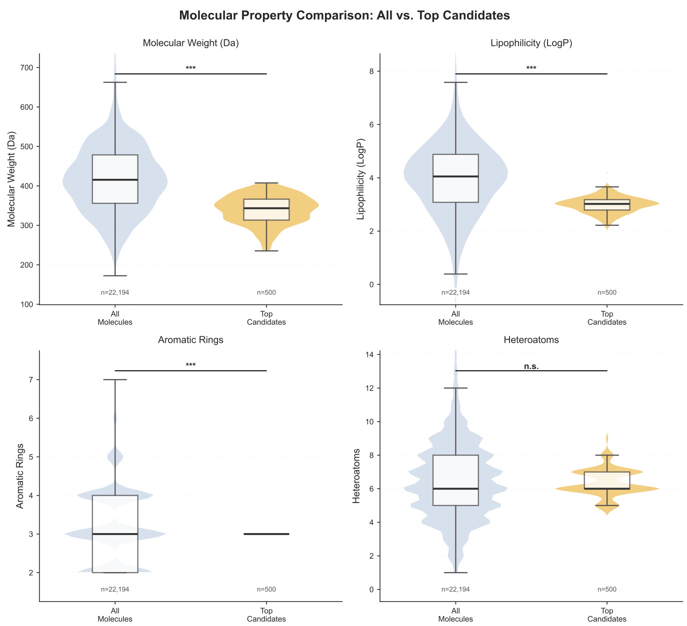

Figure S4 Property distributions of the filtered library (22 194 PubChem structures passing the donor–acceptor substructure screen, grey) versus the top 500 by heuristic TADF score (blue). Enrichment toward lower molecular weight reflects the scoring rule; these are unvalidated leads, not predicted emitters.

We additionally ranked the full library with the calibrated structure-only Morgan-FP model and attached a split-conformal 90 % prediction interval to every prediction (mean half-width 0.18 eV, empirical coverage 0.89). No ranked shortlist is deposited: for the leading candidates the predicted gap is 0.02 eV to 0.05 eV while the full interval width is about 7.6 times as large (half-width 3.8 times), and the lower bound of every one of the 100 lowest-predicted intervals falls below zero. At this accuracy the conformal procedure does not support selection at the single-molecule level, and no device-level quantity is predicted.

S6. Robustness analyses

S6.1 Cluster cross-validation

Because 92.9 % of scaffolds are singletons, we built non-singleton groups by Butina clustering (Morgan Tanimoto distance) and ran cluster-grouped CV across a cutoff scan (table S17). Even at the coarsest grouping (70 clusters, largest 38 molecules), the MAE (0.099 eV) sits between scaffold-CV (0.096 eV) and random-CV (0.106 eV), a spread of 0.010 eV. Performance is therefore *stable across validation schemes*: the headline number is not an artefact of singleton scaffolds. Regenerated by `code/cluster_cv_analysis.py`.

Table S17 Cluster-grouped CV of the headline NTO RF across Butina distance cutoffs, with scaffold and random CV for reference.

Grouping	#groups	largest	MAE (eV)	ρ
Butina 0.3	186	7	0.100	0.35
Butina 0.4	153	10	0.100	0.38
Butina 0.5	111	15	0.104	0.31
Butina 0.6	70	38	0.099	0.40
Scaffold (headline)	212	—	0.096	0.36
Random 5-fold	—	—	0.106	0.34

S6.2 Dimensionality and model capacity

Two checks address the charge that a 2 048-bit fingerprint overparameterises a 231-molecule set. First, shrinking the fingerprint to 1 024 or 512 bits leaves the MAE and the (negative) lower R^2 confidence bound essentially unchanged (table S18, top), so the negative R^2 tail is intrinsic to the low variance-explained on this target, not a width artefact; the lowest-dimensional regularised model (NTO + ElasticNet) brings the lower bound closest to zero (-0.03) but does not lift it. Second, no higher- or different-capacity learner beats the random forest under the same scaffold CV (table S18, bottom): gradient boosting, an MLP and an SVR are all worse. Model capacity is not the limiting factor. Retraining on the 208 molecules whose inter-report standard deviation is $\leq 0.05\,\mathrm{eV}$ does not lower the MAE (0.106 eV). That subset is a weak test, because 129 of those 208 molecules have no replicate at all and enter with SD $= 0$ by construction, so only 23 molecules are actually excluded. Conditioning instead on molecules that carry replicates, and reading the error off the same out-of-fold predictions, the model achieves — with a constant predictor scoring 0.074, 0.107 and 0.110 eV on the three strata against the model's 0.074, 0.106 and 0.107 eV, so the comparison carries no information about label quality — 0.074 eV on the 79 whose reports agree to $\leq 0.05\,\mathrm{eV}$, 0.106 eV on the 23 whose reports disagree, and 0.107 eV on the 129 whose label precision cannot be measured. The ordering is the one a label-limited account predicts, but the clean-versus-discordant difference is not resolved at this sample size (95 % CI -0.015 to 0.088); we report it as suggestive only. Regenerated by `code/dimensionality_check.py` and `code/capacity_and_noise_ceiling.py`.

Table S18 Top: fingerprint-width sweep (RF) — MAE and 95 % bootstrap R^2 CI. Bottom: alternative learners under the same scaffold CV. None beats the RF.

Setting	MAE (eV)	R^2 (95% CI)
Morgan 512 (RF)	0.098	0.15 (-0.44, 0.46)
Morgan 1024 (RF)	0.096	0.24 (-0.25, 0.51)
Morgan 2048 (RF)	0.091	0.27 (-0.19, 0.53)
NTO + ElasticNet	0.102	0.10 (-0.03, 0.17)
RandomForest (headline)	0.096	—
HistGradientBoosting	0.105	—
MLP (100, 50)	0.169	—
SVR (RBF)	0.120	—
ElasticNet	0.102	—

S6.3 SHAP cross-check by permutation importance

An independent, model-agnostic permutation-importance analysis (held-out 30 %, 50 repeats) ranks the singlet hole–electron overlap $S_{\mathrm{he}}(S_1)$ *first*, in agreement with TreeSHAP; two of the SHAP top-five recur in the permutation top-five. The leading attribution is thus corroborated by two methods, while the lower-ranked tail is not — consistent with our restriction of mechanistic reading to the dominant descriptor. Regenerated by `code/permutation_importance_check.py`.

S6.4 Enrichment curve and calibrated operating points

The main-text enrichment curve reports precision (fraction of selected molecules with experimental gap ≤ 0.10 eV) against screening budget. Enrichment peaks at 1.37× over the 49 % base rate around the top 5–20% and does not improve at more aggressive cuts (the top 1% is two molecules). A split-conformal calibration gives a 90 % interval half-width of 0.178 eV (empirical coverage 0.892), so a laboratory can pick an operating point at a stated confidence. Regenerated by `code/enrichment_curve.py`.

S6.5 Multi-criteria triage

A small S_1–T_1 gap is necessary but not sufficient for efficient delayed fluorescence: the emitter must also be optically bright and free of a near-degenerate second triplet. We therefore composed the predicted gap with two auxiliary structure-level filters—oscillator strength $f_{S_1} \geq 0.01$ (209/231 pass; `code/oscillator_strength_model.py`) and sTDA T_1–$T_2 > 0.10$ eV (table S14)—leaving 131 molecules that clear all three (`code/multi_criteria_triage.py`). Ranking within this pool raises the top-5 % precision for low-gap emitters from 0.58 (gap alone) to 0.71, i.e. enrichment 1.18→1.45×; the two curves converge by the top 50 %. The gain is real but small and rests on only a handful of molecules at the sharpest cut, so we read the multi-criteria filter not as a precision gain but as a physically motivated way to discard the 43 % of low-gap candidates that are dark or manifold-unsafe—a decision a scalar-gap triage cannot make. The T_1–T_2 term inherits the qualitative-only caveat of table S14.

S6.6 Dataset expansion stress-test

To test whether the benchmark is limited by corpus size, we resolved and featurised 624 additional donor–acceptor emitters from the experimental literature (name-to-structure resolution followed by the identical GFN2-xTB/sTDA natural-transition-orbital pipeline; `code/expand_dataset.py`, `code/featurize_nto_local.py`). After discarding 127 structures already present in the training set, 497 genuinely new molecules extend the corpus to 728 (table S19). Enlarging the set does *not* improve structure-only prediction: under the same scaffold CV the MAE rises from 0.098 eV to 0.124 eV and R^2 falls to −0.01 (the curated-set MAE of 0.098 eV reproduces the headline 0.096 eV within single-seed variation). Restricting the new molecules to those with replicate measurements (234 with ≥ 2 independent reports) recovers part of the loss (MAE 0.110 eV, R^2 0.05), placing one component of the degradation in label noise from automated literature extraction. We do not decompose the degradation further: the added molecules are resolved through a different structure pipeline, so label quality and chemical diversity are confounded and this design cannot apportion the loss between them. What the expansion rules out is that the corpus is merely too small. Regenerated by `code/expansion_stress_test.py`.

Table S19 Scaffold-CV of the NTO random forest as the corpus is expanded and as new-label confidence is tightened. Expansion does not improve accuracy.

Corpus	n	MAE (eV)	R^2	ρ
Curated baseline	231	0.098	0.18	0.34
+ all new	728	0.124	−0.01	0.16
+ multi-report new (≥ 2)	465	0.110	0.05	0.24

S7. Data and code availability

All data (experimental corpus, descriptor tables, predictions), trained models, and analysis/figure scripts are openly available on Zenodo (DOI: 10.5281/zenodo.17436069) under a CC-BY-4.0 licence. The primary 231-molecule benchmark descriptors were generated with `xtb` 6.7.1 (S1.1); the 497-molecule expansion stress test (S6.6) was re-featurised with `xtb` 6.7.1. Other software: `xtb4stda` with `stda` 1.6.3, Multiwfn 3.8, ORCA 6.1.1, Python 3.12, scikit-learn 1.8, RDKit 2026.03, and SHAP 0.50. Model training and cross-validation used a fixed random seed (`random_state` = 0).

S8. Inverted singlet–triplet emitters (out of scope)

The benchmark is restricted to conventional TADF emitters with a positive gap ($\Delta E_{\mathrm{ST}} > 0$). The inverted singlet–triplet (INVEST) class[12] lies outside the present pipeline: GFN2-xTB and sTDA-xTB enforce Aufbau ordering, so the random forest cannot predict a negative gap from the current feature set. A screen of the 25 482-record literature database (`code/ist_screening.py`) returns 66 reports of an inverted gap across 61 compound names (deposited, `results/ist_candidates.csv`); five of those compounds also appear in our 231-molecule corpus. For two of them the consensus median label is itself negative (HAP-3MF, −0.22 eV; TPA-NZP, −0.038 eV) and these are the only two negative labels in the corpus; the remaining three (DPXZ-DPPM, 2CzPN, UGH3) carry positive medians despite an inverted report existing. The random forest predicts a positive gap for all five, so it assigns the wrong sign to both molecules whose own training label is negative. The limitation is representational rather than statistical. A regression forest can in principle return a negative value, since its prediction is an average over training labels and two of ours are negative; what the feature sets lack is any variable that distinguishes an inverted-gap molecule, because the semi-empirical descriptors are computed under Aufbau ordering and the structure-only descriptors carry no excited-state information. More data of the same kind would not change this. INVEST screening needs ROKS- or EOM-CCSD-level descriptors[13] and is out of scope here.

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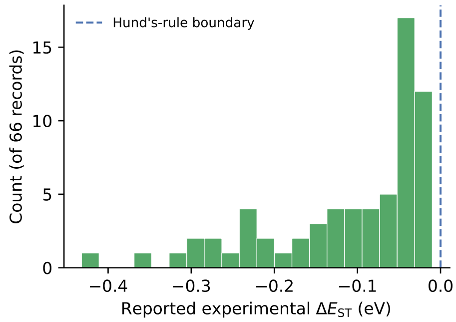

Figure S5 Reported experimental ΔE_{ST} for the 66 inverted-gap records identified in the literature database. All lie below the Hund's-rule boundary (dashed line). Structures were not resolved for these records ($\texttt{n\_with\_smiles} = 0$), so no model prediction is made for them; they are shown to indicate the reported range of the class, not as a test set.

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